

7.1.2. Let $X = \{1, 3, 5\}$ and $Y = \{a, b, c, d\}$. Define $g : X \rightarrow Y$ by the following arrow diagram.

X	g	Y
1	$\rightarrow$	b
3	$\rightarrow$	b
5	$\rightarrow$	b

Suppose X is the domain and Y is the co-domain for function g .

We must define $g : X \rightarrow Y$

g satisfies two properties:

- (1) Every element in X is related to some element in Y .
- (2) No element in X is related to more than one element in Y .

X is the domain and Y is the co-domain.

- domain of $g = \{1, 3, 5\}$
- co-domain of $g = \{a, b, c, d\}$

$$g(1) = b, g(3) = b, \text{ and } g(5) = b$$

$$\text{Range of } g = \text{image of } X \text{ under } g = \{y \in Y \mid g(x) = y \text{ for some } x \in X\}$$

Hence, range of $g = \{b\}$

Solution: $g(1) = b, g(3) = b, g(5) = b$

Domain of $g = \{1, 3, 5\}$

Co-domain of $g = \{a, b, c, d\}$

Range of $g = \{b\}$